

Tut-8 Intro to filters

1. An FIR filter of length 5 is defined by a symmetric impulse response. Let the I/P to the filter be a sum of three cosine sequences of angular freq: 0.2 rad/samples , 0.5 rad/samples , and 0.8 rad/samples , respectively. Determine the IR coeff. so that the filter passes only the mid frequency component of I/P.

2. Two well known mathematical integration formulas can be represented by difference eqⁿs as follows

$$y(n) = y(n-1) + \frac{1}{2} \{x(n) + x(n-1)\} \quad \text{with } y(-1) = 0$$

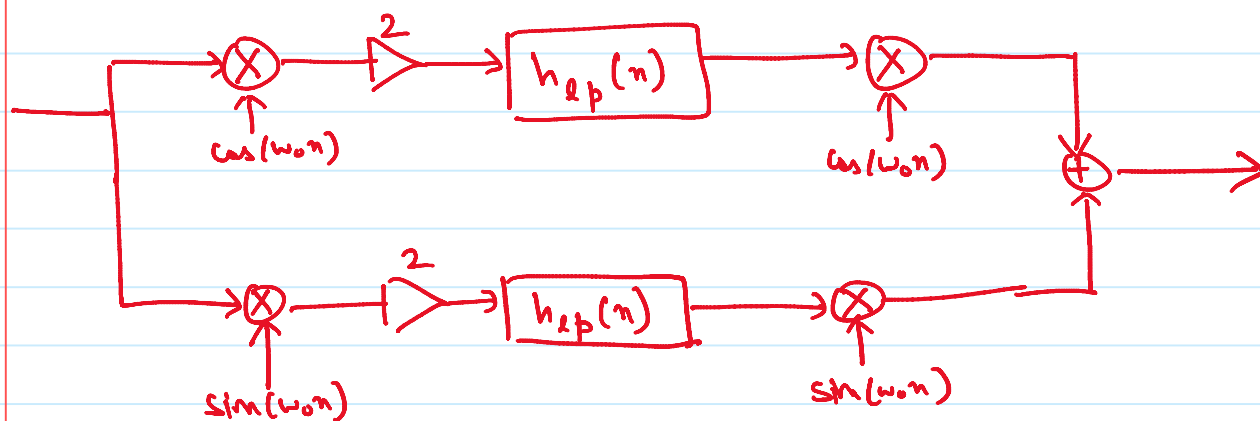
[Trapezoidal Integral formula] $\rightarrow$ IIR digital filter.

$$y(n) = y(n-2) + \frac{1}{3} \{x(n) + 4x(n-1) + x(n-2)\}$$

[Simpson's numerical integration formula] $\rightarrow$

Find the freq. response of above two filters & compare.

3. Show that the given structure represents BPF



6. $H_{LP}(z)$ be TF of an ideal real coeff LPF having cut off freq. ω_c with $\omega_c < \pi/2$. Consider random

(b) $H_{LP}(z)$ be TF of an ideal real coeff LPF having cut off freq. ω_p with $\omega_p < \pi/2$. Consider complex coeff. T.F. $H_{LP}(e^{j\omega_0} z)$ where $\omega_p < \omega_0 < \pi - \omega_p$.

Now, denote TF $G(z) = H_{LP}(e^{j\omega_0} z) + H_{LP}(e^{-j\omega_0} z)$.

S.T. $G(z)$ is a real coeff BPF with passband centered at ω_0 . Determine the width of its passband in terms of ω_p & its IR $g(n)$ in terms of IR $h_{LP}(n)$ of parent LPF.

NOTE: part (a) implements BPF asked in part (b)

4. A length 13 type 3 real coeff FIR filter has foll. zeros:

$$z_1 = 0.3 + j0.5, \quad z_2 = j0.8, \quad z_3 = -0.3.$$

- (a) Determine location of remaining zeros.
- (b) TF $H_3(z)$ of filter.

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